

$$sx(0^+) + x'(0^+) + 3x(0^+) = 2s + (-3) + 3(2)$$

L.T soln of linear differential eqn  $\rightarrow 2s - 3 + 6 = 2s + 3.$

eg 11 Solve the diff. eqn:-

$$x'' + 3x' + 2x = 0, \quad x(0^+) = 2, \quad x'(0^+) = -3.$$

Soln  $\rightarrow$

Taking Laplace transform.

$$s^2 X(s) - sx(0^+) - x'(0^+) + 3sX(s) - 3x(0^+) + 2X(s) = 0$$

Putting initial cond<sup>n</sup>,  
 $(s^2 + 3s + 2) X(s) = sx(0^+) + x'(0^+) + 3x(0^+)$   
 $(s^2 + 3s + 2) X(s) = 2s + 3.$

$$X(s) = \frac{2s+3}{s^2+3s+2} = \frac{2s+3}{(s+1)(s+2)}$$

Hence, all roots of denominator polynomial are simple.  
 Then by partial fraction expansion,

$$X(s) = \frac{2s+3}{(s+1)(s+2)} = \frac{k_1}{s+1} + \frac{k_2}{s+2}$$

Putting  $(s+1) = 0 \Rightarrow s = -1,$

$$2s+3 = k_1(s+2) + k_2(s+1)$$

$$2s+3 = k_1(s+2)$$

$$2(-1)+3 = k_1(-1+2)$$

$$\boxed{k_1 = 1}$$

Putting  $(s+2) = 0 \Rightarrow s = -2$

$$\frac{2s+3}{(s+1)(s+2)} = \frac{k_1(s+2) + k_2(s+1)}{(s+1)(s+2)}$$

$$2s+3 = k_2(s+1)$$

$$\boxed{k_2 = 1}$$

$$X(s) = \frac{k_1}{s+1} + \frac{k_2}{s+2} = \frac{1}{s+1} + \frac{1}{s+2}$$

$\therefore$  soln of given diff. eqn.

$$x(t) = \mathcal{L}^{-1}[X(s)] = \mathcal{L}^{-1}\left[\frac{1}{s+1} + \frac{1}{s+2}\right]$$

$$\boxed{x(t) = e^{-t} + e^{-2t}} \quad \underline{\text{Ans}}$$

eg 2.11: If  $I(s) = \frac{s^2 + 5s + 9}{s^3 + 5s^2 + 12s + 8}$ , find  $i(t) \Rightarrow \frac{s^2 + 5s + 9}{(s+1)(s^2 + 4s + 8)}$

Soln:

$$I(s) = \frac{s^2 + 5s + 9}{(s+1)[(s+2)^2 - (j2)^2]}$$

$$= \frac{s^2 + 5s + 9}{(s+1)(s+2+j2)(s+2-j2)}$$

$$I(s) = \frac{k_1}{(s+1)} + \frac{k_2}{(s+2+j2)} + \frac{k_3}{(s+2-j2)}$$

$$\frac{s^2 + 5s + 9}{(s+1)(s+2+j2)(s+2-j2)} = \frac{k_1(s+2+j2)(s+2-j2) + k_2(s+1)(s+2-j2) + k_3(s+1)(s+2+j2)}{(s+1)(s+2+j2)(s+2-j2)}$$

$$s = -1$$

Putting  $s = -1$ ,

$$s^2 + 5s + 9 = k_1(-1+2+j2)(-1+2-j2)$$

$\Downarrow$

$$\boxed{k = 1}$$

$$\therefore k_2 = -\frac{1}{j4}, \quad k_3 = \frac{1}{j4}$$

$$I(s) = \frac{1}{s+1} - \frac{1/j4}{(s+2+j2)} - \frac{1/j4}{(s+2-j2)}$$

$$i(t) = e^{-t} - \left(\frac{1}{j4}\right) e^{-(2+j2)t} + \left(\frac{1}{j4}\right) e^{-(2-j2)t}$$

$$= e^{-t} - \frac{1}{j4} e^{-2t} [e^{-j2t} - e^{j2t}]$$

$$i(t) = e^{-t} + \frac{1}{2} e^{-2t} \sin 2t$$

$$I(s) = \frac{1}{s+1} + \frac{1}{s^2+4s+8}$$

$$= \frac{1}{s+1} + \frac{1}{2} \left[ \frac{2}{(s+2)^2 + (2)^2} \right]$$

$$c(t) = \mathcal{L}^{-1}[I(s)] = e^{-t} + \frac{1}{2} e^{-2t} \sin 2t \quad \underline{\text{Ans}}$$

$$\left[ \begin{smallmatrix} \infty \\ 0 \end{smallmatrix} \mathcal{L}^{-1} \left[ \frac{\omega}{(s+a)^2 + \omega^2} \right] = e^{-at} \sin \omega t \right]$$

$$\frac{8(5s+3)}{8(s+1)} = \underline{3}$$

$$e^{-at} \sin \omega t$$

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$$= \frac{\omega}{(s+a)^2 + \omega^2}$$

$$\Rightarrow e^{-2t} \sin 2t$$

$$= \frac{2}{(s+2)^2 + (2)^2}$$

Q31.

Consider the diff eqn

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = 5 U(t)$$

$$+ 2y(t) = 5 U(t) \rightarrow \text{unit step fun}$$

$$\text{L.T} = 5U(T) = \frac{5}{s}$$

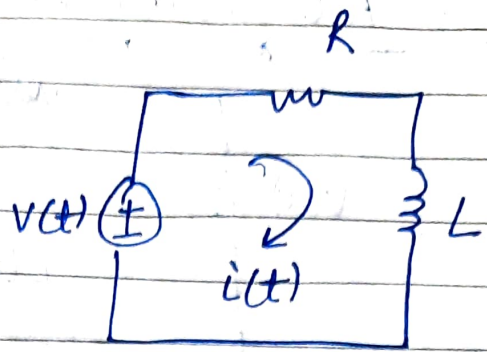
$$y(0) = -1, \quad \frac{dy}{dt}(0^+) = 2, \quad \text{Determine } y(t)$$

Soln:

$$y(t) = \frac{5}{2} - 5e^{-t} + \frac{3}{2} e^{-2t} \left[ y(s) = \frac{5}{2s} + \frac{-5}{s+1} + \frac{3}{2(s+2)} \right]$$



① R-L circuit 1-



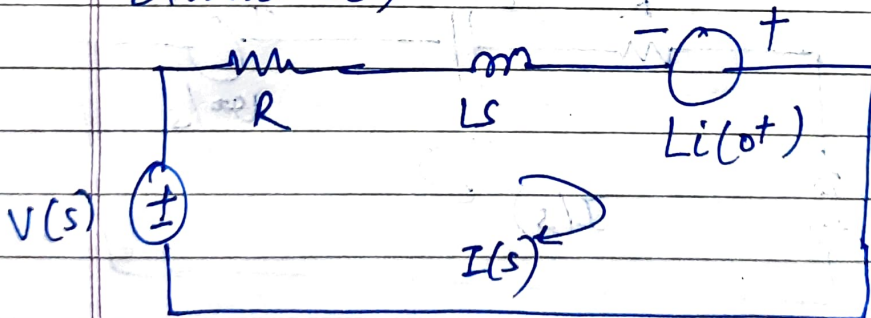
$$Ri(t) + L \frac{di(t)}{dt} = V(t)$$

$$\text{As } \frac{di(t)}{dt} = [sI(s) - i(0^+)]$$

Taking L.T,

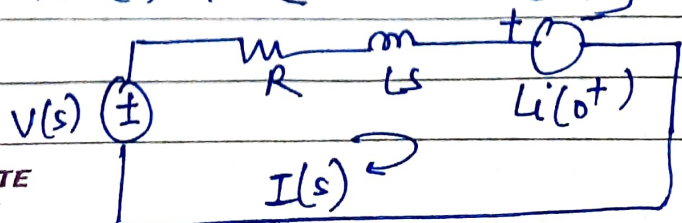
$$RI(s) + L[sI(s) - i(0^+)] = V(s)$$

Case I:- we assume that initial cond<sup>n</sup> i.e. if the initial current  $[i(0^+)]$  is clockwise, the initial current assist the charging current  $i(t)$



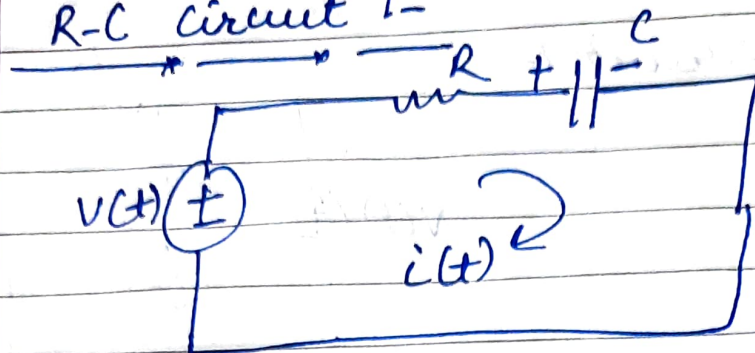
Case II:- If the initial current is anti-clockwise  $[-i(0^+)]$  i.e. the initial current opposes the charging current then voltage becomes

$$V(s) = RI(s) + L[sI(s) + i(0^+)]$$



②

R-C circuit 1-

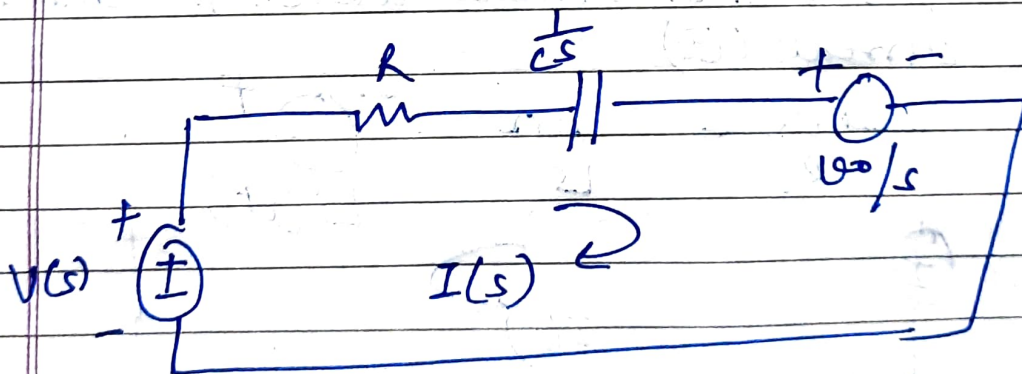


$$V(t) = Ri(t) + \frac{1}{C} \int_0^{\infty} i(t) dt + v_0$$

Taking L-T,

$$V(s) = \left[ RI(s) + \frac{I(s)}{Cs} + \frac{v_0}{s} \right]$$

$$V(s) = RI(s) + \frac{I(s)}{Cs} + \frac{v_0}{Cs}$$

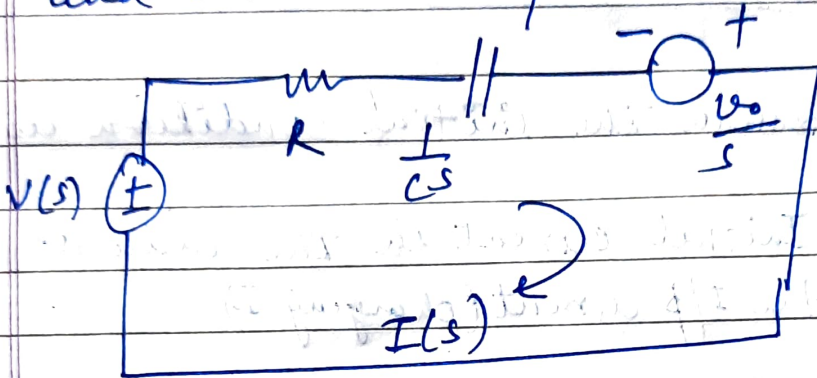


If the initial voltage is  $v_0$  with the +ve polarity at the side of the capacitor then it opposes the charging current  $i(t)$ .

If the polarity of the initial voltage is  $-V_0$  i.e. it assists the charging current  $i(t)$  then

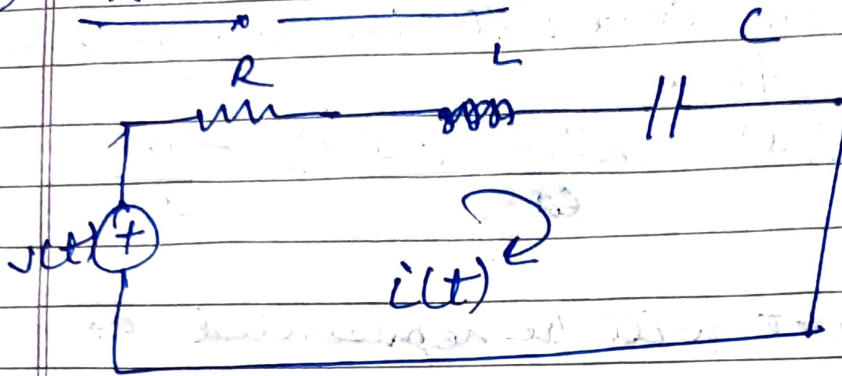
$$V(s) = \frac{RI(s)}{s} + \frac{I(s)}{Cs} - \frac{V_0}{s}$$

and ckt will be represented as



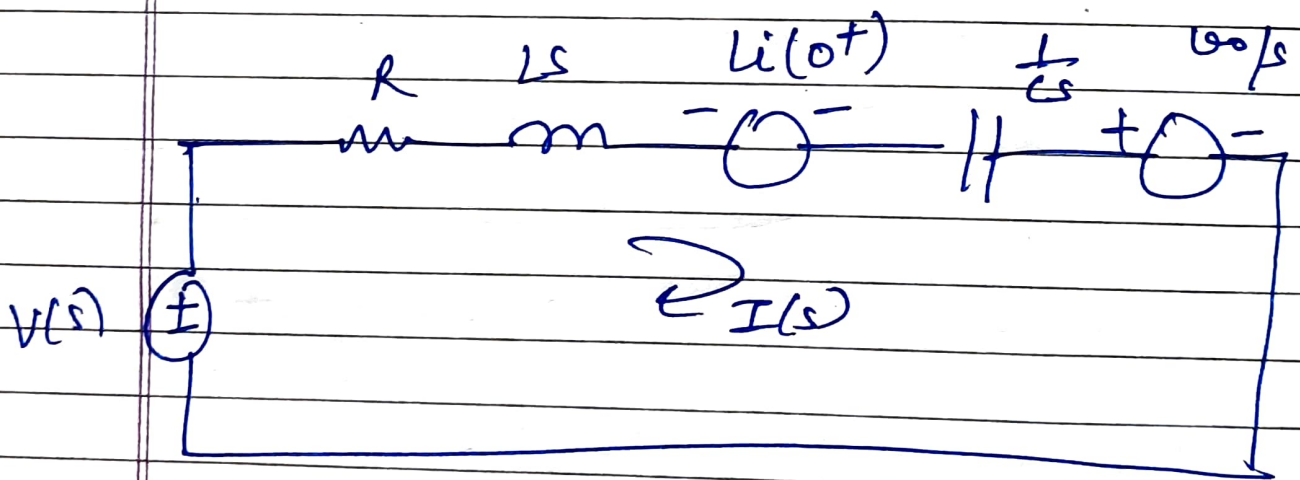


### ③ R-L-C Circuit :-



Assume the initial condition as follows:-

- ① Initial current in the inductor existing the I/p current (charging I)
- ② Initial capacitor voltage opposes the I/p current (charging I)



$$V(t) = Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int_0^t i(t) dt + v_0(0^+)$$

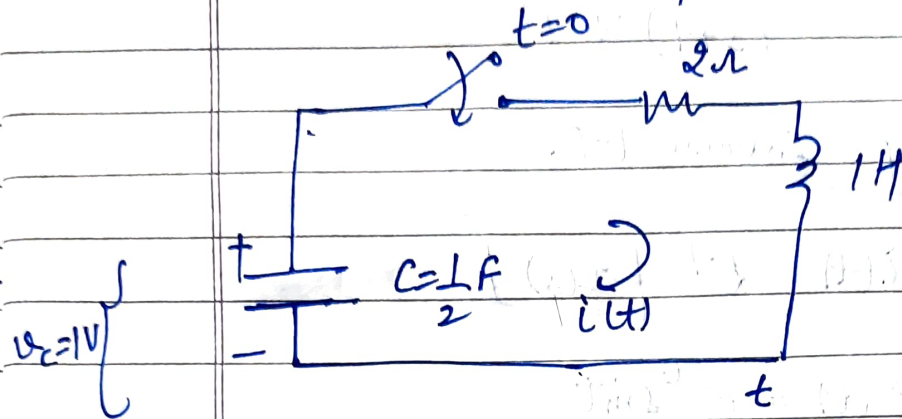
Taking L-T,

$$V(s) = RI(s) + L[sI(s) - i(0^+)] + \frac{I(s)}{Cs} + \frac{v_0(0^+)}{s}$$



eg:-

Consider a series R-L-C circuit with The capacitor initially charged to 'voltage of 1V'. find the expression for  $i(t)$ . Also draw s-domain representation of the ckt.



Soln:-  $Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int_0^t i(t) dt + V_C(0^+) = 0$

Taking L.T,

$$RI(s) + L[sI(s) - i(0^+)] + \frac{1}{Cs} I(s) + \frac{V_C(0^+)}{s} = 0$$

$$V_C(0^+) = -1 \text{ (with given polarity)}$$

$$RI(s) + LsI(s) + \underbrace{Li(0^+)}_0 + \frac{I(s)}{Cs} + \frac{V_C(0^+)}{s} = 0$$

$$2I(s) + sI(s) + \frac{2}{s} I(s) - \frac{1}{s} = 0$$

$$I(s) \left( 2 + s + \frac{2}{s} \right) = \frac{1}{s}$$

$$I(s) = \frac{1}{s^2 + 2s + 2}$$

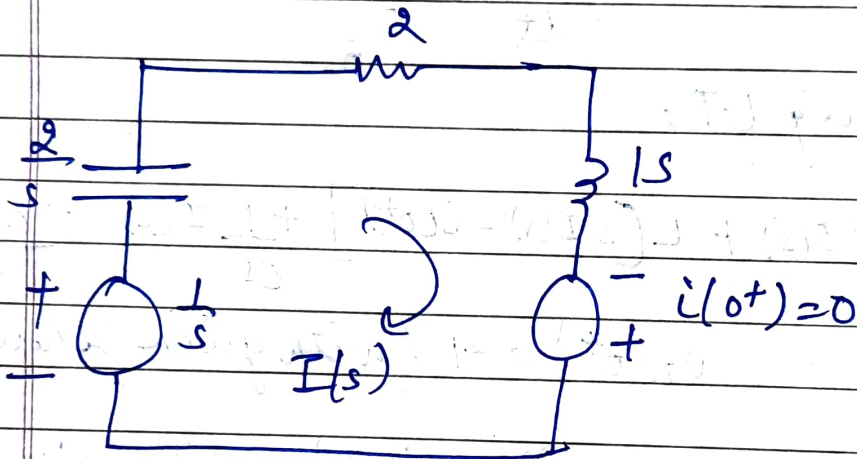
$$I(s) = \frac{1}{s^2 + 2s + 2}$$

$$I(s) = \frac{1}{(s+1)^2 + 1^2}$$

Taking Inverse L-T,

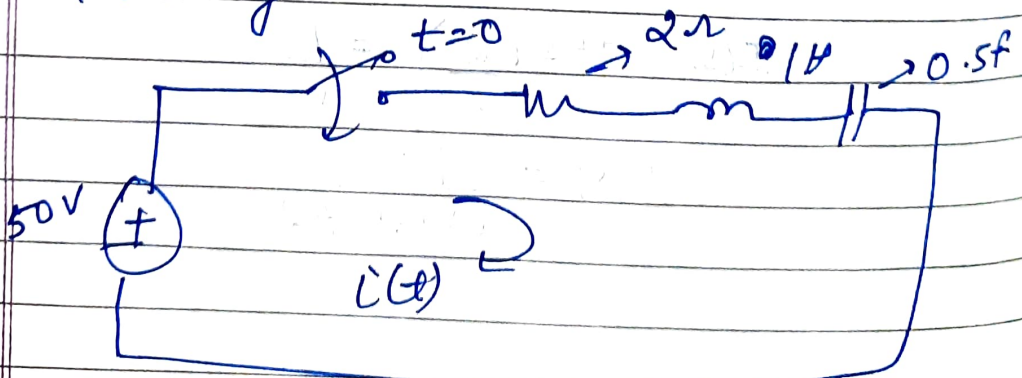
$$i(t) = \mathcal{L}^{-1} [I(s)]$$

$$i(t) = e^{-t} \sin t$$



eg 21-

In series RLC ckt, there is no initial charge on the Capacitor. If the switch S is closed at  $t=0$ , Determine the resulting  $I$ .



Soln<sup>1</sup> -  $Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int_0^t i(t) dt + q_c(0^+) = V(t)$   
 $\Downarrow$   
 no initial charge = 0

Taking L-T,

$$RI(s) + LS I(s) - Li(0^+) + \frac{I(s)}{Cs} = \frac{V(s)}{s}$$

Putting all values,

$$2I(s) + 1s I(s) - i(0^+) + \frac{I(s)}{0.5s} = \frac{50}{s}$$

no initial current,  $i(0^+) = 0$

$$2I(s) + s I(s) + \frac{I(s)}{0.5s} = \frac{50}{s}$$

$$I(s) \left( 2 + s + \frac{1}{0.5s} \right) = \frac{50}{s}$$

$$I(s) = \frac{50}{s^2 + 2s + 2} = \frac{50}{(s+1)^2 + 1^2}$$

Taking Inverse L.T,

$$i(t) = \mathcal{L}^{-1} [I(s)]$$

$$i(t) = 50 e^{-t} \sin t \quad \checkmark$$



**EXAMPLE 5.7** Consider the  $R$ - $C$  parallel circuit as shown in figure 5.7(a) with  $R = 0.5\Omega$  and  $C = 4F$  excited by d.c. current source of 10 A. Determine the voltage across the capacitor by applying Laplace transformation. Assume the initial voltage across the capacitor as 2V. Also draw the  $s$ -domain representation of the circuit.

**Solution :** Applying KCL,

$$\frac{v(t)}{R} + C \frac{dv(t)}{dt} = 10$$

Taking Laplace transform

$$\frac{V(s)}{R} + C[sV(s) - v(0^+)] = \frac{10}{s}$$

$$2V(s) + 4[sV(s) - 2] = \frac{10}{s}$$

$$\text{or} \quad V(s) = \frac{8s + 10}{s(4s + 2)}$$

Applying the partial fraction expansion,

$$V(s) = \frac{2s + 2.5}{s(s + 0.5)} = \frac{K_1}{s} + \frac{K_2}{s + 0.5}$$

$$K_1 = sV(s)|_{s=0} = 5$$

$$\text{and} \quad K_2 = (s + 0.5)V(s)|_{s=-0.5} = -3$$

$$\text{or} \quad V(s) = \frac{5}{s} + \frac{-3}{s + 0.5}$$

$$\text{Therefore, } v(t) = \mathcal{L}^{-1}[V(s)] = 5 - 3e^{-0.5t} \text{ V}$$

The  $s$ -domain representation of the circuit is shown in figure 5.7 (b).

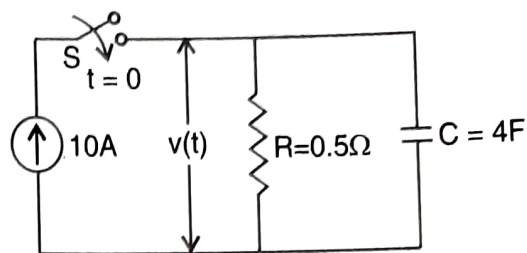


Fig. 5.7(a).

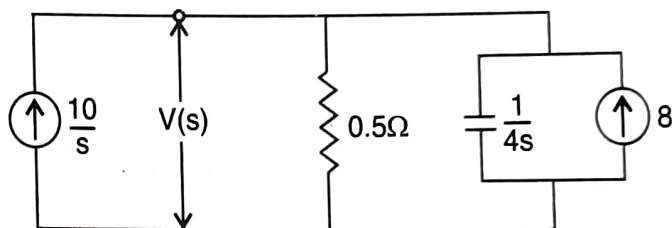


Fig. 5.7(b).

**EXAMPLE 5.8** In the network shown in figure 5.8, the switch  $S$  is closed at  $t = 0$ . With the network parameter values shown, find the expressions for  $i_1(t)$  and  $i_2(t)$ , if the network is unenergized before the switch is closed.

**Solution :** Applying KVL, Loop equations are

$$\frac{di_1(t)}{dt} + 10i_1(t) + 10[i_1(t) - i_2(t)] = 100$$

$$\text{or} \quad \frac{di_1(t)}{dt} + 20i_1(t) - 10i_2(t) = 100 \quad \dots(i)$$

$$\text{And, } \frac{di_2(t)}{dt} + 10i_2(t) + 10[i_2(t) - i_1(t)] = 0$$

$$\text{or} \quad \frac{di_2(t)}{dt} + 20i_2(t) - 10i_1(t) = 0 \quad \dots(ii)$$

The transform equations of (i) and (ii) may be written as (keeping initial conditions are zero, as given) :

$$(s + 20) I_1(s) - 10I_2(s) = \frac{100}{s}$$

$$\text{and} \quad -10I_1(s) + (s + 20) I_2(s) = 0$$

Writing in matrix form,

$$\begin{bmatrix} s + 20 & -10 \\ -10 & s + 20 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} \frac{100}{s} \\ 0 \end{bmatrix}$$

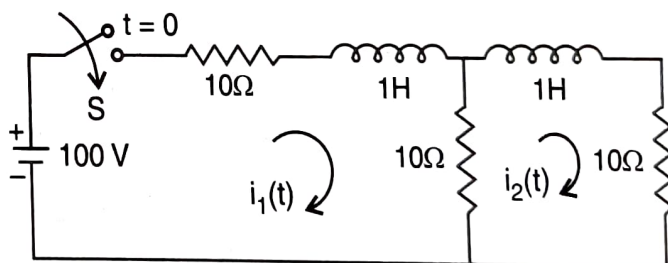


Fig. 5.8.

By Cramer's rule,

$$I_1(s) = \frac{\Delta_1}{\Delta} \text{ and } I_2(s) = \frac{\Delta_2}{\Delta}$$

Where,

$$\Delta_1 = \begin{vmatrix} 100 & -10 \\ s & s+20 \end{vmatrix} ; \Delta_2 = \begin{vmatrix} s+20 & 100 \\ -10 & s \end{vmatrix} ; \text{ and } \Delta = \begin{vmatrix} s+20 & -10 \\ -10 & s+20 \end{vmatrix}$$

Therefore,

$$\Delta_1 = \frac{100}{s}(s+20) ; \Delta_2 = \frac{1000}{s}$$

and

$$\Delta = (s+20)^2 - 100 = s^2 + 40s + 300$$

So,

$$I_1(s) = \frac{100(s+20)}{s(s^2 + 40s + 300)}$$

and

$$I_2(s) = \frac{1000}{s(s^2 + 40s + 300)}$$

The partial fraction expansion of above expressions  $I_1(s)$  and  $I_2(s)$  are

$$I_1(s) = \frac{20/3}{s} + \frac{-5}{s+10} + \frac{-5/3}{s+30}$$

and

$$I_2(s) = \frac{10/3}{s} + \frac{-5}{s+10} + \frac{5/3}{s+30}$$

The inverse Laplace transformation give  $i_1(t)$  and  $i_2(t)$  as

$$i_1(t) = \frac{20}{3} - 5e^{-10t} - \frac{5}{3}e^{-30t} \text{ A}$$

$$i_2(t) = \frac{10}{3} - 5e^{-10t} + \frac{5}{3}e^{-30t} \text{ A}$$

Which is the required solution

**EXAMPLE 5.13** Solve for  $i(t)$  in circuit as shown in figure 5.10 (a) in which 3F capacitor is initially charged to 20V, the 6-F capacitor to 10V, and the switch is closed at  $t = 0$ . Also draw the trans-formed circuit.

**Solution :**

$$\frac{1}{3} \int_{-\infty}^t i(t) dt + 1i(t) + 1i(t) + \frac{1}{6} \int_{-\infty}^t i(t) dt = 0$$

$$\text{or } \frac{1}{3} \int_0^t i(t) dt - 20 + 2i(t) + \frac{1}{6} \int_0^t i(t) dt + 10 = 0$$

Taking Laplace transform

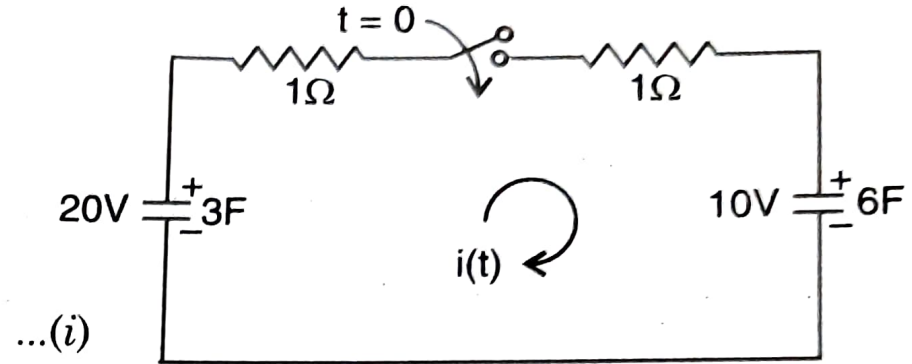
$$\frac{1}{3s} I(s) - \frac{20}{s} + 2I(s) + \frac{1}{6s} I(s) + \frac{10}{s} = 0 \quad \dots(ii)$$

$$\text{or } \left( \frac{1}{3s} + \frac{1}{6s} \right) I(s) + 2I(s) = \frac{10}{s}$$

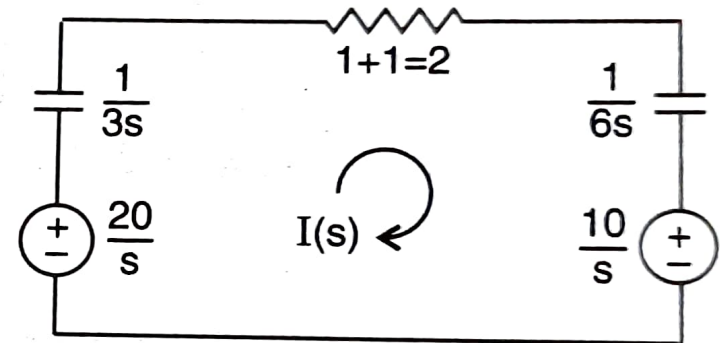
$$\text{or } \frac{1}{2s} I(s) + 2I(s) = \frac{10}{s}$$

$$\text{or } I(s) = \frac{20}{4s+1} = \frac{5}{s+0.25}$$

Therefore,  $i(t) = \mathcal{L}^{-1}[I(s)] = 5 e^{-0.25 t} \text{ A}$



**Fig. 5.10(a).**



**Fig. 5.10(b).**



**EXAMPLE 5.16** In the circuit shown in figure 5.13(a), switch  $S$  is in position 1 for a long time and moved to position 2 at  $t = 0$ . Find the voltage across the capacitor  $v_c(t)$  for  $t > 0$ .

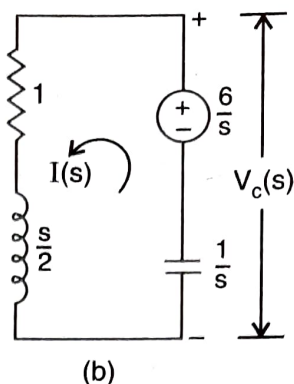
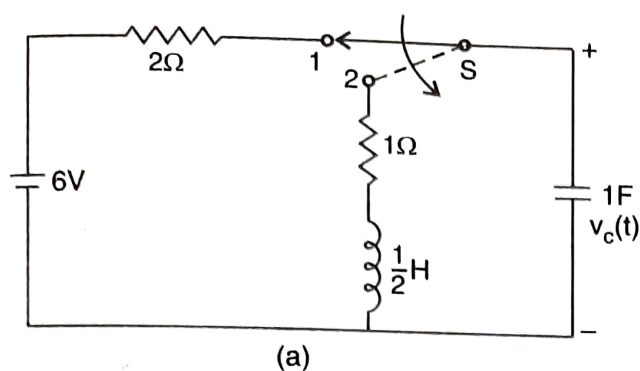


Fig. 5.13.

**Solution :** When the  $S$  is in position 1 for a long time, capacitor charges up to 6V. i.e.

$$v_c(0^+) = 6$$

Now when the  $S$  is in position 2 for  $t > 0$ . Applying KVL, in the transformed circuit is shown in figure 5.13(b).

The current  $I(s)$  in the loop is given by

$$I(s) \left[ 1 + \frac{s}{2} + \frac{1}{s} \right] = \frac{6}{s}$$

$$I(s) = \frac{12}{s^2 + 2s + 2}$$

And

$$\begin{aligned} V_c(s) &= I(s) \cdot \left[ 1 + \frac{s}{2} \right] \quad \text{or} \quad \frac{6}{s} - I(s) \cdot \frac{1}{s} \\ &= \frac{6(s+2)}{s^2 + 2s + 2} = \frac{6(s+1)}{(s+1)^2 + 1} + \frac{6}{(s+1)^2 + 1} \end{aligned}$$

Therefore,  $v_c(t) = \mathcal{L}^{-1}[V_c(s)] = 6e^{-t} \cos t + 6e^{-t} \sin t = 6e^{-t} (\cos t + \sin t) \text{ V}$

**EXAMPLE 5.17** In the circuit shown in figure 5.14 steady state is reached with  $S$  open.  $S$  is closed at  $t = 0$ . Determine  $i(t)$  and  $v(t)$  for  $t > 0$ .

**Solution :** At  $t = 0^-$ , as steady state is reached  $L$  is short circuited and  $C$  is open circuited.

Hence 
$$i(0^-) = \frac{25}{2 + \frac{1}{2}} = 10 \text{ A}$$

and 
$$v(0^-) = 10 \times \frac{1}{2} = 5 \text{ V}$$

Now after  $S$  is closed. Applying KVL,

$$2i(t) + 1 \cdot \frac{di(t)}{dt} + v(t) = 0$$

Taking Laplace transform

$$(2 + s)I(s) - i(0^+) + V(s) = 0$$

or 
$$(2 + s)I(s) - 10 + V(s) = 0$$

$$(\text{As } i(0^+) = i(0^-) = 10\text{A})$$

or 
$$(2 + s)I(s) + V(s) = 10$$

...(i)

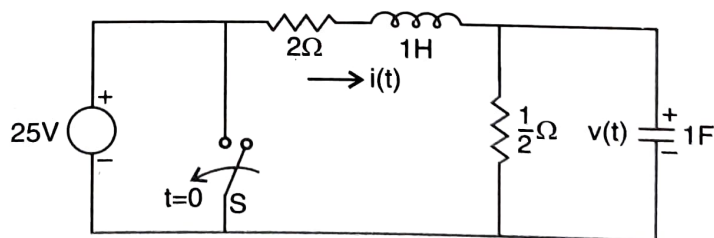


Fig. 5.14.

And, 
$$i(t) = \frac{v(t)}{1/2} + 1 \cdot \frac{dv(t)}{dt}$$

Taking Laplace transform,

$$I(s) = 2 V(s) + sV(s) - v(0^+)$$

$$I(s) = (2 + s) V(s) - 5$$

$$(\text{As } v(0^+) = v(0^-) = 5V)$$

$$-I(s) + (2 + s) V(s) = 5$$

Writing equations (i) and (ii) in Matrix form,

$$\begin{bmatrix} 2+s & 1 \\ -1 & 2+s \end{bmatrix} \begin{bmatrix} I(s) \\ V(s) \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

By Cramer's Rule,

$$I(s) = \frac{\Delta_1}{\Delta} \quad \text{and} \quad V(s) = \frac{\Delta_2}{\Delta}$$

Where

$$\Delta = \begin{vmatrix} 2+s & 1 \\ -1 & 2+s \end{vmatrix} = (2+s)^2 + 1$$

$$\Delta_1 = \begin{vmatrix} 10 & 1 \\ 5 & 2+s \end{vmatrix} = 20 + 10s - 5 = 10s + 15$$

and

$$\Delta_2 = \begin{vmatrix} 2+s & 10 \\ -1 & 5 \end{vmatrix} = 10 + 5s + 10 = 5s + 20$$

$$I(s) = \frac{10s+15}{(s+2)^2+1} = \frac{10(s+2)}{(s+2)^2+1} - \frac{5}{(s+2)^2+1}$$

Therefore,

$$i(t) = \mathcal{L}^{-1}[I(s)] = 10 e^{-2t} \cos t - 5 e^{-2t} \sin t \quad \text{A}$$

And

$$V(s) = \frac{5s+20}{(s+2)^2+1} = \frac{5(s+2)}{(s+2)^2+1} + \frac{10}{(s+2)^2+1}$$

therefore,

$$\begin{aligned} v(t) &= \mathcal{L}^{-1}[V(s)] \\ &= 5 e^{-2t} \cos t + 10 e^{-2t} \sin t \quad \text{A} \end{aligned}$$

**EXAMPLE 5.22** In the series  $R$ - $C$  circuit of figure 5.19, the capacitor has initial charge  $2.5 \text{ mC}$ . At  $t = 0$ , the switch is closed and a constant voltage source  $V = 100 \text{ V}$  is applied, find  $i(t)$ .

**Solution :** Applying KVL,

$$100 = 10 i(t) + \frac{1}{50 \times 10^{-6}} \int_0^t i(t) dt + v_c(0^+)$$

Taking Laplace transform, with  $v_c(0^+)$

$$= \frac{-Q_0}{C} = \frac{-2.5 \times 10^{-3}}{50 \times 10^{-6}}$$

$$= -50 \text{ V} \quad (\text{with given polarity})$$

$$\frac{100}{s} = 10I(s) + \frac{1}{50 \times 10^{-6}} \cdot \frac{I(s)}{s} - \frac{50}{s}$$

$$I(s) \left[ 10 + \frac{1}{(50 \times 10^{-6})s} \right] = \frac{150}{s}$$

$$I(s) \left[ \frac{10s + (2 \times 10^4)}{s} \right] = \frac{150}{s}$$

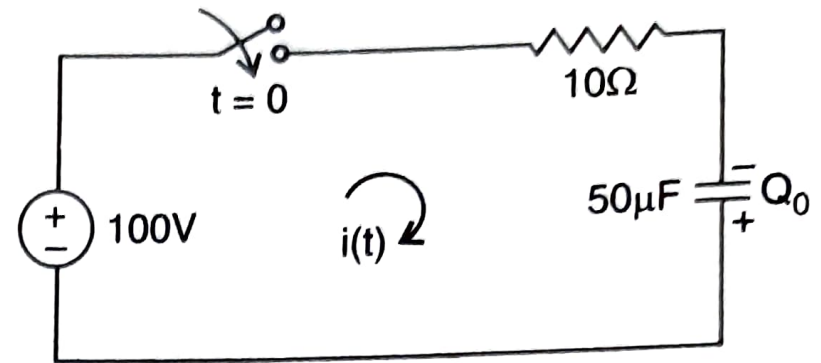
$$I(s) = \frac{150}{10s + (2 \times 10^4)} = \frac{15}{s + (2 \times 10^3)}$$

Therefore,

$$i(t) = \mathcal{L}^{-1}[I(s)] = 15e^{-2 \times 10^3 t}$$

or

$$i(t) = 15 e^{-2t} \text{ A; where } t \text{ in msec.}$$



**Fig. 5.19.**

**EXAMPLE 5.34** In the circuit of the figure 5.28, the switch  $S$  is closed at  $t = 0$  with the capacitor initially unenergised. For the numerical values given, find  $i(t)$ .

**Solution :** Applying KVL,

$$10 e^{-t} \sin t = 10 i(t) + \left\{ \frac{1}{10 \times 10^{-6}} \int_0^t i(t) dt + 0 \right\}$$

Taking laplace transform,

$$\frac{10}{(s+1)^2 + 1} = 10 I(s) + 10 \times 10^4 \frac{I(s)}{s}$$

$$I(s) = \frac{10 s}{\{(s+1)^2 + 1\} (10 s + 10 \times 10^4)} = \frac{s}{\{(s+1)^2 + 1\} (s + 10^4)}$$

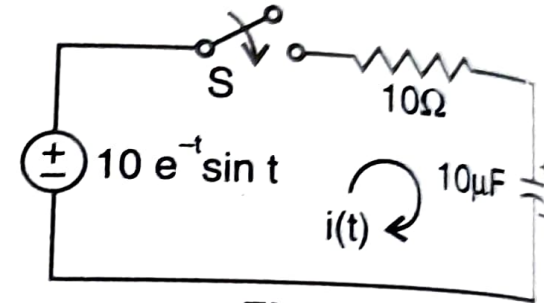
Using partial fraction expansion, we have

$$I(s) = \frac{As + B}{(s+1)^2 + 1} + \frac{C}{s + 10^4} \cong \frac{10^{-4} s}{(s+1)^2 + 1} - \frac{10^{-4}}{s + 10^4}$$

$$I(s) = 10^{-4} \left[ \frac{s+1}{(s+1)^2 + 1} - \frac{1}{(s+1)^2 + 1} \right] - 10^{-4} \frac{1}{s + 10^4}$$

Taking inverse Laplace transform, we have

$$i(t) = 10^{-4} e^{-t} (\cos t - \sin t) - 10^{-4} e^{-10^4 t}.$$



**Fig. 5.28.**